



```
In [23]: import nltk
        from nltk.tokenize import sent_tokenize
        from nltk.tokenize import word_tokenize
        from nltk.corpus import stopwords
        from nltk.stem import PorterStemmer
        from nltk.stem import WordNetLemmatizer
        import re
        from sklearn.feature_extraction.text import TfidfVectorizer
```

```
In [24]: nltk.download('punkt')
        nltk.download('stopwords')
        nltk.download('wordnet')
        nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to C:\Users\Pratik
[nltk_data]   Pawar\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to C:\Users\Pratik
[nltk_data]   Pawar\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to C:\Users\Pratik
[nltk_data]   Pawar\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]   C:\Users\Pratik Pawar\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]   date!
```

```
Out[24]: True
```

```
In [25]: text = """India is a vast country with the second highest population in the world with diverse cultures, traditions and beliefs. People in India celebrate unity in diversity. Festivals like Diwali, Holi, Navratri, Ramzan, Christmas etc. are celebrated by people across India and create a sense of brotherhood and cultural unity. Each festival has its religious and cultural importance.
```

```
In [26]: sentences = sent_tokenize(text)
        print(sentences)
```

```
['India is a vast country with the second highest population in the world.', 'It is a country\nwith diverse cultures, traditions and beliefs.', 'People in India celebrate unity in diversity.', 'Festivals like Diwali, Holi, Navratri, Ramzan, Christmas etc.', 'are celebrated by people across India\nand create a sense of brotherhood and cultural unity.', 'Each festival has its religious and cultural importance.']
```

```
In [27]: words = word_tokenize(text)
        print(words)
```

```
['India', 'is', 'a', 'vast', 'country', 'with', 'the', 'second', 'highest', 'po-
pulation', 'in', 'the', 'world', '.', 'It', 'is', 'a', 'country', 'with', 'diver-
se', 'cultures', ',', 'traditions', 'and', 'beliefs', '.', 'People', 'in', 'In-
dia', 'celebrate', 'unity', 'in', 'diversity', '.', 'Festivals', 'like', 'Diwal-
i', ',', 'Holi', ',', 'Navratri', ',', 'Ramzan', ',', 'Christmas', 'etc', '.',
'are', 'celebrated', 'by', 'people', 'across', 'India', 'and', 'create', 'a',
'sense', 'of', 'brotherhood', 'and', 'cultural', 'unity', '.', 'Each', 'festiva-
l', 'has', 'its', 'religious', 'and', 'cultural', 'importance', '.']
```

```
In [28]: stop_words = set(stopwords.words('english'))
print(stop_words)
```

```
{'myself', 'couldn', "didn't", 'here', 'in', 're', 'you', 'her', "she'll", 'doe-
s', "she'd", 'into', "he'll", 'all', 'other', 'this', 'during', 'i', 'itself',
'both', "you're", 'hasn', 'have', 'theirs', 'those', 'against', 'mustn', 'm',
"mightn't", 'below', 'can', 's', 'do', 'him', 'be', 'haven', 'further', 'afte-
r', 'own', 'doing', 'no', 'how', 'she', 'before', 'up', "i'm", 'then', 'wasn',
"you'd", "doesn't", 'me', 'there', 've', 'why', 'more', 'shan', 'because', "i-
t's", 'his', "we're", "they'd", 'between', 'they', 'than', 'mightn', 'off', 'yo-
urs', 'having', 'their', 'out', "we've", 'only', 'from', 'll', 'for', "we'd",
"i'll", 'above', "won't", "wouldn't", 't', 'a', 'don', 'ourselves', 'where', "s-
han't", 'herself', 'these', 'so', 'just', 'but', 'has', "don't", 'isn', 'few',
'any', 'o', 'we', "weren't", 'of', 'while', "you'll", "aren't", "he's", 'its',
"hadn't", 'or', "we'll", 'which', 'now', "hasn't", 'over', 'until', 'some', 'ha-
dn', "needn't", 'being', 'again', 'each', 'an', 'were', 'your', "i'd", 'when',
'won', 'same', 'about', "you've", "it'll", 'once', 'what', 'too', 'he', 'them',
'such', 'isn't', 'ma', 'under', 'd', "should've", "that'll", "they'll", "could-
n't", 'our', 'themselves', 'hers', 'wouldn', 'who', 'weren', "they're", 'was',
'himself', "mustn't", 'through', 'should', "shouldn't", 'that', 'y', 'by', 'doe-
sn', 'on', 'my', 'nor', 'will', 'ain', 'are', 'am', 'and', "he'd", 'ours', 'ver-
y', 'with', "they've", "wasn't", "she's", 'is', "i've", 'at', 'not', 'yourself',
"haven't", 'if', 'most', 'shouldn', 'the', 'needn', 'to', 'down', 'had', 'a-
ren', 'didn', "it'd", 'did', 'as', 'yourselves', 'it', 'whom', 'been'}
```

```
In [29]: text="How to remove stop words with NLTK library in Python7"
text= re.sub('[^a-zA-Z]', ' ', text)
tokens= word_tokenize(text.lower())
filtered_text= []
for w in tokens:
    if w not in stop_words:
        filtered_text.append(w)
print("Tokenized Sentence:", tokens)
print("Filtered Sentence:", filtered_text)
```

```
Tokenized Sentence: ['how', 'to', 'remove', 'stop', 'words', 'with', 'nltk', 'l-
ibrary', 'in', 'python']
```

```
Filtered Sentence: ['remove', 'stop', 'words', 'nltk', 'library', 'python']
```

```
In [30]: ps = PorterStemmer()
stemmed_words = ['wait', 'waiting', 'waited', 'waits']
for w in stemmed_words:
    print(w, ":", ps.stem(w))
print("Original Sentence:", stemmed_words)
```

```
wait : wait
waiting : wait
waited : wait
waits : wait
Original Sentence: ['wait', 'waiting', 'waited', 'waits']
```

```
In [31]: lemmatizer = WordNetLemmatizer()
text= "studies studying cies cry"
tokenization= nltk.word_tokenize(text)
for w in tokenization:
    print("Lemma for {} is {}".format(w, lemmatizer.lemmatize(w)))
```

```
Lemma for studies is study
Lemma for studying is studying
Lemma for cies is cies
Lemma for cry is cry
```

```
In [32]: data = 'The pink sweater fits her perfectly'
words = nltk.word_tokenize(data)
for word in words:
    print(nltk.pos_tag([word]))
```

```
[('The', 'DT')]
[('pink', 'NN')]
[('sweater', 'NN')]
[('fits', 'NNS')]
[('her', 'PRP$')]
[('perfectly', 'RB')]
```

```
In [33]: paragraph= """India is a vast country with second highest populati
on in the world. It is a country
with diverse cultures, traditions and beliefs. People in India cel
ebrate unity in diversity.
Festivals like Diwali, Holi, Navratri, Ramzan, Christmas etc. are
celebrated by people across India
and create a sense of brotherhood and cultural unity. Each festiva
l has its religious and cultural importance."""
```

```
In [34]: wn = WordNetLemmatizer()
sentences = nltk.sent_tokenize(paragraph)
corpus = []
for i in range(len(sentences)):
    review = re.sub('[^a-zA-Z]', ' ', sentences[i])
    review = review.lower()
    review = review.split()
    review = [wn.lemmatize(word) for word in review if not word in set(stopwor
review = ' '.join(review)
    corpus.append(review)
print(corpus)
```

```
['india vast country second highest populati world', 'country diverse culture t
radition belief', 'people india cel ebrate unity diversity', 'festival like diw
ali holi navratri ramzan christmas etc', 'celebrated people across india create
sense brotherhood cultural unity', 'festiva l religious cultural importance']
```

```
In [35]: tfidf = TfidfVectorizer()
X = tfidf.fit_transform(corpus).toarray()
print(X)
```

```
[[0.         0.         0.         0.         0.         0.
  0.33061545 0.         0.         0.         0.         0.
  0.         0.         0.         0.         0.         0.40318254
  0.         0.         0.27912828 0.         0.         0.
  0.40318254 0.         0.         0.40318254 0.         0.
  0.         0.40318254 0.40318254]
 [0.         0.46262479 0.         0.         0.         0.
  0.37935895 0.         0.         0.46262479 0.46262479 0.
  0.         0.         0.         0.         0.         0.
  0.         0.         0.         0.         0.         0.
  0.         0.         0.         0.         0.         0.46262479
  0.         0.         0.         ]
 [0.         0.         0.         0.45529187 0.         0.
  0.         0.         0.         0.         0.         0.45529187
  0.         0.45529187 0.         0.         0.         0.
  0.         0.         0.31520422 0.         0.         0.37334585
  0.         0.         0.         0.         0.         0.
  0.37334585 0.         0.         ]
 [0.         0.         0.         0.         0.         0.35355339
  0.         0.         0.         0.         0.         0.
  0.35355339 0.         0.35355339 0.         0.35355339 0.
  0.35355339 0.         0.         0.35355339 0.35355339 0.
  0.         0.35355339 0.         0.         0.         0.
  0.         0.         0.         ]
 [0.36523197 0.         0.36523197 0.         0.36523197 0.
  0.         0.36523197 0.29949544 0.         0.         0.
  0.         0.         0.         0.         0.         0.
  0.         0.         0.25285463 0.         0.         0.29949544
  0.         0.         0.         0.         0.36523197 0.
  0.29949544 0.         0.         ]
 [0.         0.         0.         0.         0.         0.
  0.         0.         0.42790272 0.         0.         0.
  0.         0.         0.         0.52182349 0.         0.
  0.         0.52182349 0.         0.         0.         0.
  0.         0.         0.52182349 0.         0.         0.
  0.         0.         0.         ]]
```